

Inert Cue Scheduling and Audit

Supply time, confirm explicitly, preserve every decision

Opaque visual labels; no external action seam

Lesson	029 — component	Time	180–240 minutes
Prerequisites	Lessons 001, 002, 011, 017	Board	Arduino Mega 2560
Evidence	cue LEDs, state cadence, TP29, audit	Status	host verified; bench open
Energy class	E1: USB, passive buttons, resistor-limited LEDs		

Boundary. Cue IDs are opaque visual labels. They do not name pins, addresses, frequencies, launcher channels, or external equipment. This lesson has no relay, transistor-driven load, initiator, radio, captured protocol, generic callback, or output adapter. Confirmation is instructional redundancy, not a safety interlock.

1 Mission and evidence chain

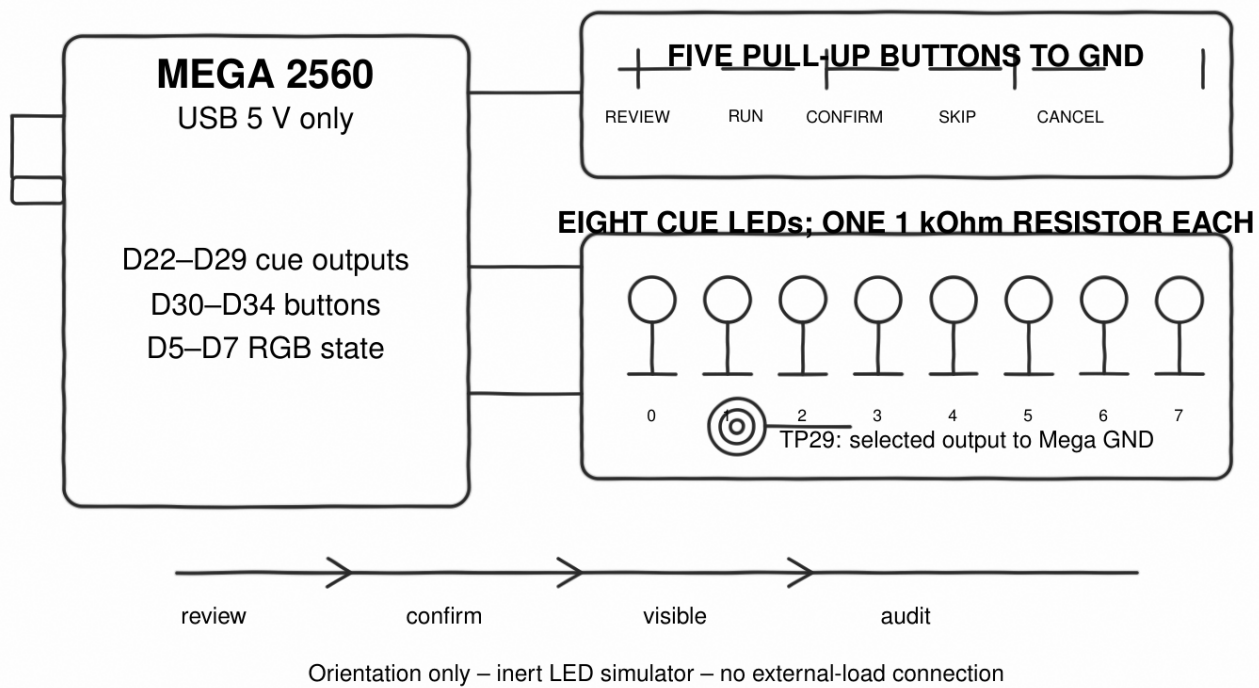
plan + supplied time + review/confirm → snapshot → inert LEDs + bounded audit.

The core receives values and returns values. Only the canonical example owns the supported low-voltage indicators.

1.1 Learning outcomes

- validate a fixed plan without allocation or a hidden clock;
- explain epoch-based offsets and confirmation-time visibility;
- distinguish Waiting, Confirmation, Active, Held, and terminal phases;
- preserve delayed decisions in an append-only audit;
- replay byte-stable records without treating Serial as proof;
- separate acquisition, active presentation, shutdown, and physical power removal.

2 Orientation and unpowered inspection



Alternative text: orientation plate showing one USB-powered Mega, five pull-up buttons on D30–D34, eight separately resistor-limited cue LEDs on D22–D29, a separately resistor-limited common-cathode RGB state LED on D5–D7, and TP29 at the selected cue output. No external-load connection is present.

Role	Mega pin	Connection
cue indicators	D22–D29	pin to 1 kΩ, LED anode; cathode to GND
operator controls	D30–D34	Review, Run, Confirm, Skip, Cancel to GND; pull-up
RGB state	D5–D7	red, green, blue through separate 1 kΩ resistors
Lesson 029 TP29	selected cue output	Mega-pin side of resistor, relative to GND

Unpowered prediction. Verify polarity, one resistor per LED channel, five switch-to-ground paths, common ground, and no connection outside the USB-powered fixture. Record resistor values: _____.

3 Current and resource worksheet

The sketch commands at most one cue channel and one RGB channel. A deliberately conservative zero-forward-voltage bound gives

$$I_{\text{channel}} \leq \frac{5.0 \text{ V}}{1000 \, \Omega} = 5.0 \text{ mA}, \quad I_{\text{LED, total}} \leq 10.0 \text{ mA}.$$

This is a design bound, not a bench measurement.

Quantity	Predicted	Measured	Calculated	Interpretation
USB/5 V rail	at most 5.0 V	_____	—	_____
selected cue resistor	1 kΩ	_____	$I = (V - V_f)/R$	_____
active RGB resistor	1 kΩ	_____	$I = (V - V_f)/R$	_____
worst simultaneous LED load	at most 10 mA	_____	_____	_____

D5–D7 use PWM-capable pins, but the example selects one fixed channel at a time. D22–D34 are ordinary digital endpoints. No bus, interrupt, radio, motor, or external supply is acquired.

4 Plan and scheduler contract

A valid plan contains 1–32 cues. IDs are unique values 0–31, offsets strictly increase, visible durations are nonzero, active intervals do not overlap, and the plan end stays inside the unambiguous half of 32-bit time.

Opaque ID	Offset	Visible for	Canonical display position
3	2000 ms	2000 ms	cue LED index 0
7	7000 ms	2000 ms	cue LED index 1
12	12000 ms	2000 ms	cue LED index 2
29	17000 ms	2000 ms	cue LED index 3

Run establishes the plan epoch. Later offsets never move when confirmation is late. A cue becomes visible at accepted confirmation and stays visible for its own duration. A delayed update atomically hides expired work, records every decision, and reaches the latest still-open cue or Complete without a burst.

5 Operator sequence and phase truth

1. Hold Review to enter Review.
2. Press Run once to establish the epoch.
3. Inspect the named next cue during Confirmation.
4. Press Confirm inside the inclusive three-second window.

Run, Confirm, Skip, and Cancel are debounced edges; Review is a level. More than one non-cancel edge is invalid. Cancel has absolute priority. Releasing Review blanks an active cue and enters Held. Review alone does not resume; press Run.

Phase	Valid intent	Visible evidence
Idle/Review	hold Review, then Run	cue off; steady state channel
Waiting	wait, cancel, or release Review	cue off; slow pulse
Confirmation	Confirm, Skip, cancel, or release	cue off; fast pulse
Active	wait, cancel, or release	exactly one cue LED
Held	hold Review and press Run	cue off; distinct slow pulse
Complete	none; terminal	cue off; steady completion state
Cancelled/Fault	none; terminal	cue off; fault state

One supplied timestamp may carry only one changed input frame. Repeating the identical frame is idempotent; changing it at the same timestamp is `InvalidArgument`. The sketch samples once per distinct `millis()` tick.

6 Bounded audit and replay

`CueAuditBuffer` uses caller-owned append storage. It never overwrites old evidence. Capacity exhaustion enters `Held` with `CapacityExceeded`; operation cannot continue unrecorded. Scheduler shutdown appends `Shutdown` and preserves records until audit-buffer shutdown clears them.

Event group	Stable tokens
lifecycle	initialized, shutdown
operator	review-started, run-requested, confirmed, held, resumed, cancelled
cue	confirmation-requested, cue-shown, cue-hidden, cue-skipped
terminal	faulted, completed

```
adk-cue,1,<sequence>,<ticks>,<event>,<cue-or-dash>,<index-or-dash>,<status>
```

The encoder is atomic, requires at most 96 bytes, includes a newline, and adds no NUL byte. Absent cue fields are `-`, `-`. Serial may carry these bytes for inspection, but is never the circuit-native observation path.

7 Narrative Mega example

```
observeOperator      (now);
decideCueSchedule    (now);
presentCueSnapshot   (now);
```

Setup acquires five buttons, eight cue LEDs, the RGB indicator, and scheduler; configures every output inactive; then starts. The loop observes one complete frame, decides once, and presents one stable snapshot. It lights a cue only when `phase == Active && hasCue`, using `cueIndex`. Opaque cue never indexes a pin.

8 Predict, observe, interpret

1. Predict the steady acquisition indication appears only after every resource and the scheduler initialize successfully.
2. Hold Review and press Run. Predict Lesson 029 TP29 remains inactive before confirmation.
3. Confirm once. Predict only the selected cue goes active for $[t_{\text{confirm}}, t_{\text{confirm}} + \text{visibleFor})$.
4. Observe TP29 and LED identity. Interpret agreement only as evidence for snapshot-to-inert-display mapping.
5. Release Review while Active. Predict immediate blanking and Held.
6. Invoke shutdown separately; record immediate darkness and inactive TP29, then remove USB power.

A dark LED and inactive voltage do not alone establish high impedance. Deterministic ownership tests separately prove resource release. Physical high-impedance evidence requires a named weak-pull fixture or instrument method; leave that result open unless it is actually performed.

9 Diagnosis and stop conditions

Observation	Action and interpretation
more than one cue LED	remove USB; inspect index mapping and wiring
cue before Confirm	remove USB; do not treat the fixture as valid evidence
Held does not blank cue	remove USB; inspect snapshot presentation
audit reaches capacity	expect Held/CapacityExceeded; do not continue
unexpected or out-of-rail TP29	remove USB; inspect ground, polarity, resistor
hot part or unstable USB	remove USB immediately; do not troubleshoot powered

10 Open hardware acceptance record

Software evidence already established: deterministic host tests, strict headers/style, Mega compile, measured size, monochrome PDF, and site package. None is a physical result.

Board/revision	_____	Commit	_____
LED specimens/polarity	_____	Resistors	_____
Supply/instrument	_____	Tool versions	_____
Operator/date	_____	Deviations	_____

- ☐ unpowered circuit, polarity, resistors, and prohibited external connections inspected;
- ☐ acquisition evidence predicted, observed, interpreted;
- ☐ TP29 confirmation and exact visibility boundary measured;
- ☐ hold, resume, timeout, skip, cancel, fault, and audit-full cases observed;
- ☐ shutdown inactive state and resource release recorded as separate claims;
- ☐ reset and power-removal outcomes recorded.

Prediction: _____

Observation: _____

Interpretation and limits: _____

Stop rule. Remove USB power for a hot part, unstable supply, wrong polarity, missing resistor, unexpected indicator, out-of-rail TP29, or schematic/breadboard disagreement. Software shutdown is lifecycle behavior, never an emergency stop.

11 Reproduce the software evidence

```
make host-test
make host-test-exceptions
make arduino-Lesson029InertCueSchedule
make size-check
make lessons
make lessons-check
make site-check
make hardware-card LESSON=029 BOARD=mega2560
```

Measured with Arduino AVR core 1.8.8 and avr-gcc 7.3.0, the canonical sketch uses 10,704 bytes of flash and 1,826 bytes of static RAM on the Mega 2560. Budgets are 24,576 and 2,048 bytes respectively. Re-measure when source or toolchain identity changes.

12 Reference trail

- ADK `inert_cue_scheduler.h`, `cue_audit.h`, focused tests, and canonical Lesson 029 sketch;
- ADK development, testing, safety, packaging, and PDF policies;
- Arduino Mega 2560 Rev3 datasheet for the exact board profile.

The searchable HTML reference is the primary accessible route. This printable PDF uses real text, headings, lists, and monochrome graphics but does not claim PDF/UA conformance.